**Experiment 2: University Database Using MongoDB**

**Aim:** To create a university database with schema validation, insert sample data, and perform aggregation queries using `$lookup`, `$match`, `$group`, `$project`, `$sort`, `$unwind`, and `$facet`.

### a) Switch to database

```
use universityDB
```

### b) Create collections with validation

```
db.createCollection("students", { validator: { $jsonSchema: { bsonType:
"object", required: ["studentId","name","age","deptId"], properties: {
studentId: { bsonType: "int" }, name: { bsonType: "string" }, age: {
bsonType: "int", minimum: 18 }, deptId: { bsonType: "int" } } } } });

db.createCollection("courses", { validator: { $jsonSchema: { bsonType:
"object", required: ["courseId","courseName","deptId"], properties: {
courseId: { bsonType: "int" }, courseName: { bsonType: "string" }, deptId:
{ bsonType: "int" } } } } });

db.createCollection("departments", { validator: { $jsonSchema: { bsonType:
"object", required: ["deptId","deptName"], properties: { deptId: {
bsonType: "int" }, deptName: { bsonType: "string" } } } } });

db.createCollection("faculty", { validator: { $jsonSchema: { bsonType:
"object", required: ["facultyId","name","deptId","joiningDate"],
properties: { facultyId: { bsonType: "int" }, name: { bsonType: "string" },
deptId: { bsonType: "int" }, joiningDate: { bsonType: "date" } } } } });

db.createCollection("enrollments", { validator: { $jsonSchema: { bsonType:
"object", required: ["studentId","courseId","enrollDate"], properties: {
studentId: { bsonType: "int" }, courseId: { bsonType: "int" }, enrollDate:
{ bsonType: "date" } } } } });
```

### c) Insert sample data (10 documents each)

### Students

```
db.students.insertMany([
 {studentId:1,name:"Arjun",age:20,deptId:101},
 {studentId:2,name:"Priya",age:21,deptId:102},
 {studentId:3,name:"Ravi",age:22,deptId:101},
 {studentId:4,name:"Meena",age:20,deptId:103},
 {studentId:5,name:"Amit",age:23,deptId:101},
 {studentId:6,name:"Kiran",age:19,deptId:102},
 {studentId:7,name:"Sita",age:20,deptId:103},
 {studentId:8,name:"Raj",age:21,deptId:104},
 {studentId:9,name:"Anita",age:22,deptId:101},
 {studentId:10,name:"Vikram",age:23,deptId:104}
]);
```

### Departments

```
db.departments.insertMany([
 {deptId:101,deptName:"Computer Science"},
```

```
 {deptId:102,deptName:"Electronics"},
 {deptId:103,deptName:"Mechanical"},
 {deptId:104,deptName:"Civil"},
 {deptId:105,deptName:"Architecture"},
 {deptId:106,deptName:"Biotechnology"},
 {deptId:107,deptName:"Mathematics"},
 {deptId:108,deptName:"Physics"},
 {deptId:109,deptName:"Chemistry"},
 {deptId:110,deptName:"English"}
]);
```

## Courses

```
db.courses.insertMany([
 {courseId:1,courseName:"DBMS",deptId:101},
 {courseId:2,courseName:"Data Structures",deptId:101},
 {courseId:3,courseName:"Circuits",deptId:102},
 {courseId:4,courseName:"Thermodynamics",deptId:103},
 {courseId:5,courseName:"Surveying",deptId:104},
 {courseId:6,courseName:"Microbiology",deptId:106},
 {courseId:7,courseName:"Calculus",deptId:107},
 {courseId:8,courseName:"Optics",deptId:108},
 {courseId:9,courseName:"Organic Chemistry",deptId:109},
 {courseId:10,courseName:"Literature",deptId:110}
]);
```

## Faculty

```
db.faculty.insertMany([
 {facultyId:1,name:"Dr. Kumar",deptId:101,joiningDate:ISODate("2018-06-
10")},
 {facultyId:2,name:"Dr. Meera",deptId:102,joiningDate:ISODate("2017-07-
15")},
 {facultyId:3,name:"Dr. Ramesh",deptId:103,joiningDate:ISODate("2019-03-
20")},
 {facultyId:4,name:"Dr. Anita",deptId:104,joiningDate:ISODate("2016-08-
25")},
 {facultyId:5,name:"Dr. Vijay",deptId:105,joiningDate:ISODate("2020-01-
12")},
 {facultyId:6,name:"Dr. Sunita",deptId:106,joiningDate:ISODate("2021-09-
05")},
 {facultyId:7,name:"Dr. Karthik",deptId:107,joiningDate:ISODate("2015-11-
11")},
 {facultyId:8,name:"Dr. Rekha",deptId:108,joiningDate:ISODate("2019-02-
14")},
 {facultyId:9,name:"Dr. Arvind",deptId:109,joiningDate:ISODate("2020-05-
18")},
 {facultyId:10,name:"Dr. Nisha",deptId:110,joiningDate:ISODate("2022-04-
21")}
]);
```

## Enrollments

```
db.enrollments.insertMany([
 {studentId:1,courseId:1,enrollDate:ISODate("2023-08-01")},
 {studentId:1,courseId:2,enrollDate:ISODate("2023-08-02")},
 {studentId:2,courseId:3,enrollDate:ISODate("2023-08-03")},
 {studentId:3,courseId:1,enrollDate:ISODate("2023-08-04")},
 {studentId:4,courseId:4,enrollDate:ISODate("2023-08-05")},
```

```
 {studentId:5,courseId:2,enrollDate:ISODate("2023-08-06")},
 {studentId:6,courseId:3,enrollDate:ISODate("2023-08-07")},
 {studentId:7,courseId:4,enrollDate:ISODate("2023-08-08")},
 {studentId:8,courseId:5,enrollDate:ISODate("2023-08-09")},
 {studentId:9,courseId:1,enrollDate:ISODate("2023-08-10")}
]);
```

### d) Aggregation queries

### i) List each student with department name

```
db.students.aggregate([
 { $lookup: { from:"departments", localField:"deptId",
foreignField:"deptId", as:"deptInfo" } },
 { $unwind:"$deptInfo" },
 { $project: { _id:0, studentId:1, name:1, department:"$deptInfo.deptName"
} }
]);
```

### ii) Show each student's enrolled courses with course names

```
db.students.aggregate([
 { $lookup: { from:"enrollments", localField:"studentId",
foreignField:"studentId", as:"enrollments" } },
 { $unwind:"$enrollments" },
 { $lookup: { from:"courses", localField:"enrollments.courseId",
foreignField:"courseId", as:"courseInfo" } },
 { $unwind:"$courseInfo" },
 { $project: { _id:0, name:1, course:"$courseInfo.courseName",
enrollDate:"$enrollments.enrollDate" } }
]);
```

### iii) Number of students per department

```
db.students.aggregate([
 { $group: { _id:"$deptId", totalStudents:{ $sum:1 } } },
 { $lookup: { from:"departments", localField:"_id", foreignField:"deptId",
as:"deptInfo" } },
 { $unwind:"$deptInfo" },
 { $project: { _id:0, department:"$deptInfo.deptName", totalStudents:1 } },
 { $sort: { totalStudents:-1 } }
]);
```

### iv) Faculty list with their courses and departments

```
db.faculty.aggregate([
 { $lookup: { from:"departments", localField:"deptId",
foreignField:"deptId", as:"deptInfo" } },
 { $unwind:"$deptInfo" },
 { $lookup: { from:"courses", localField:"deptId", foreignField:"deptId",
as:"coursesTaught" } },
 { $project: { name:1, department:"$deptInfo.deptName",
coursesTaught:"$coursesTaught.courseName" } }
]);
```

### v) Department summary: Total students, faculty and courses

```
db.departments.aggregate([
 { $lookup: { from:"students", localField:"deptId", foreignField:"deptId",
as:"studentsList" } },
 { $lookup: { from:"faculty", localField:"deptId", foreignField:"deptId",
as:"facultyList" } },
 { $lookup: { from:"courses", localField:"deptId", foreignField:"deptId",
as:"coursesList" } },
 { $project: { _id:0, deptName:1, totalStudents:{ $size:"$studentsList" },
totalFaculty:{ $size:"$facultyList" }, totalCourses:{ $size:"$coursesList"
} } }
]);
```

**Result:** The university database was created with schema validation and sample data. Aggregation queries were performed to generate student, course, faculty, and department reports.